

LESSON 303 (1.5 AATD)

Lesson Objective

During this lesson, the student will review the full panel instrument maneuvers listed. In addition, the student will be introduced to partial panel attitude instrument flying with emphasis on the additional interpretation of instruments and aircraft control under partial panel.

Briefing

- Review homework assigned in the previous lesson.
- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Student questions.

Lesson Content (Simulator)

System Power Up

- Student Tells, Student Does.
 - Have the student start the simulator using the checklist and assist.
 - Ensure that the student **checks GPS database currency.**
 - **Half-left, half-up, flag out of view.**
 - Ensure that there are no red 'x's.
 - Have the student start the engine using the checklist.

Instrument Cockpit Check

- Student Tells, Student Does.
 - Have the student read each item from the checklist out loud.
 - Demonstrate to the student what needs to be done for each item.
 - Discipline the student to set **bugs, bearing pointers**, and any **auxiliary** features.

Before Taxi Checks

- Student Tells, Student Does.
 - Airspeed indicator (PFD and standby): zero.
 - Attitude indicator (PFD): pitch/bank deviations +/- 5 degrees.
 - Standby attitude indicator: discuss standby battery check and adjust the miniature aircraft (cannot be performed in the ATD).
 - Altimeter (PFD and standby): +/- 75 feet of station elevation.
 - Vertical speed indicator (PFD): reads 0 or use current indication.
 - Trim coordinator: wings level or no vector when still; inclinometer centered when still; skids when turning.
 - HSI: matches compass and turns appropriately.
 - Magnetic compass: indicates heading properly, no bubbles, card present, and swings freely.
 - Use a logical flow while checking instruments!
 - Discuss the proper actions to take if a piece of equipment fails its check.

Give the student a departure via radar vectors. The instructor simulates ATC. Have the student switch to the appropriate frequencies throughout the simulator lesson.

Basic Instrument Maneuvers - Full Panel

- Straight and level:
 - Student Tells, Student Does.
 - Remind the student to begin level-off **10% vertical speed.**
 - Assist the student with pitch and power settings for the simulator.
 - Allow the student to get a feel for **trimming.**
 - Training cruise for the appropriate aircraft is to be maintained unless otherwise specified.
 - The appropriate **checklists** must be completed.
- Pattern F:
 - Instructor Tells, Student Does.
 - Ensure level-offs are $\frac{1}{2}$ **bank angle**; level-offs should begin **10% VSI.**
 - Coach the student on **changing angles of bank with airspeed changes,** and pitch moment changes with flaps - Monitor accuracy of turns and timekeeping.
 - Upon completion of the maneuver, compare the student's time to the required time.
- Vertical S-1:
 - Instructor Tells, Student Does.
 - The instructor will call out the climbs, descents, and keeps time.
 - Time and altitude will indicate the accuracy of flying – accurate flying is shown by reaching target altitudes at 1:00.
 - Monitor the flying and coach use of pitch and power – level offs should begin **10% VSI.**
 - Once complete, note the time.
- Steep turns:
 - Student Does, Instructor Evaluates.
 - Note heading – conduct **clearing turns at 30° bank preceding** the maneuver.
 - Coach the student in holding the correct pitch and power setting.
 - Scan will need to be rapid to remain within parameters.
 - It is best to alternate directions for proficiency.
- Recognition and recovery from unusual attitudes:
 - Student Does, Instructor Evaluates.
 - It is recommended to conduct at least 2, and they may be done at random.
 - This is knowledge from private pilot training, so the student should be able to recover promptly and correctly.

Preceding partial panel training, it is recommended to fail the student's PFD or ADAHRS. Have the student use the Abnormal Checklist.

Basic Instrument Maneuvers - Partial Panel

- Pattern D:
 - Instructor Tells, Student Does.
 - In this maneuver, the student will complete all turns through the use of timing.
 - Coach the student on the **revised instrument scan** - note that there is still an attitude indicator.
 - Make note of the differences **between the simulator and the airplane**.
 - Evaluate the accuracy of each turn - rollouts should be made as soon as time expires.
 - Upon completion, note the time.
- Compass turns:
 - Instructor Tells, Student Does.
 - Either prior to, in conjunction with, or succeeding the partial panel patterns, compass turns are to be made.
 - Start with cardinal headings to introduce the concept of **UNOS**.
 - If needed, have a diagram of compass overshoot/undershoot degrees.

If there is spare time in the lesson, conduct additional patterns or have the student practice compass and timed turns.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 304.
- Assign Homework:
 - Instrument Flying Handbook:
 - Chapter 07.
 - Chapter 08.

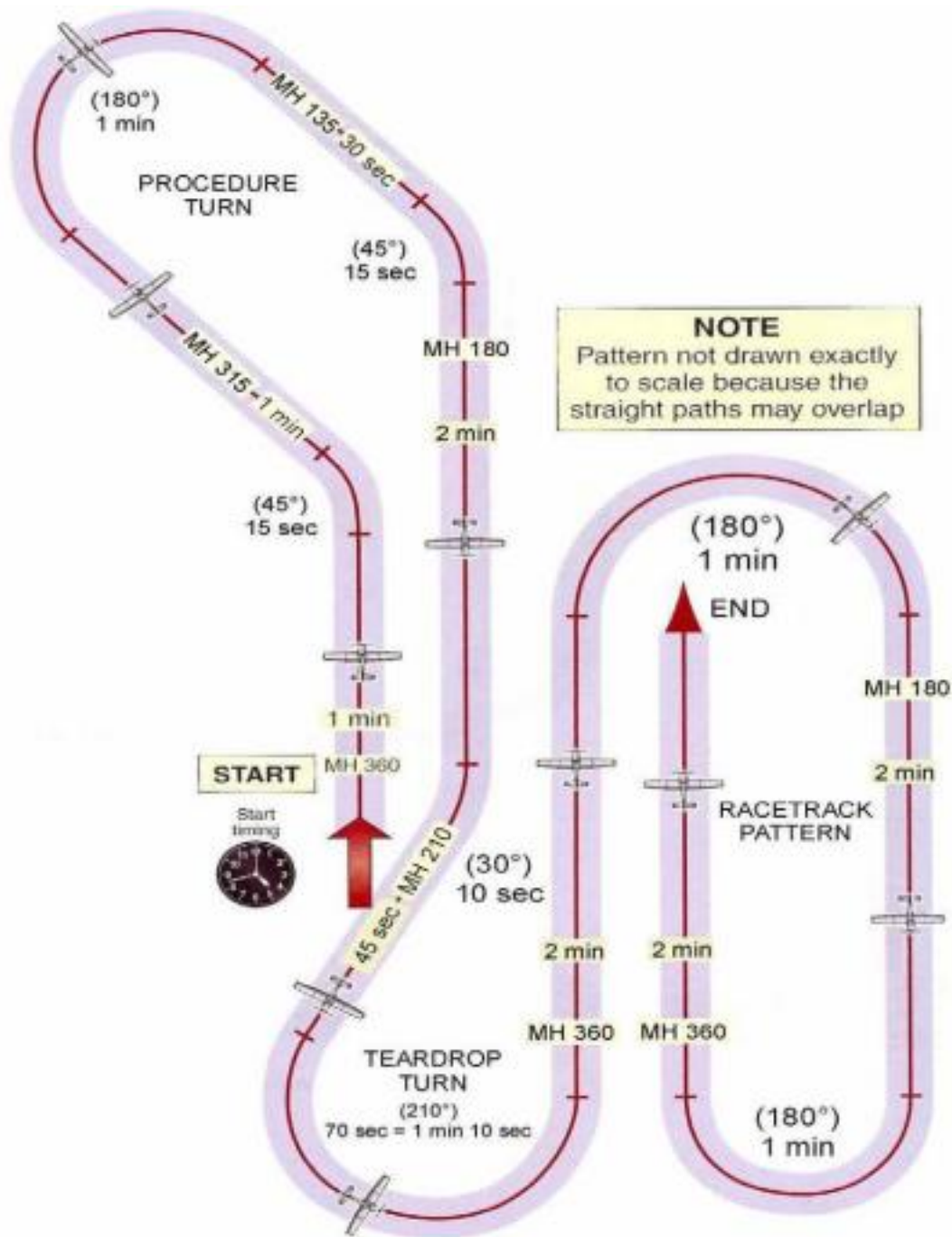
Completion Standards

This lesson will be complete when the student can, with minimal instructor assistance:

1. Demonstrate aircraft control through the use of pitch, bank and power.
2. Altitude should be maintained within + 200 feet, headings within + 15 degrees and airspeeds within + 15 knots.

Instrument Rating Course

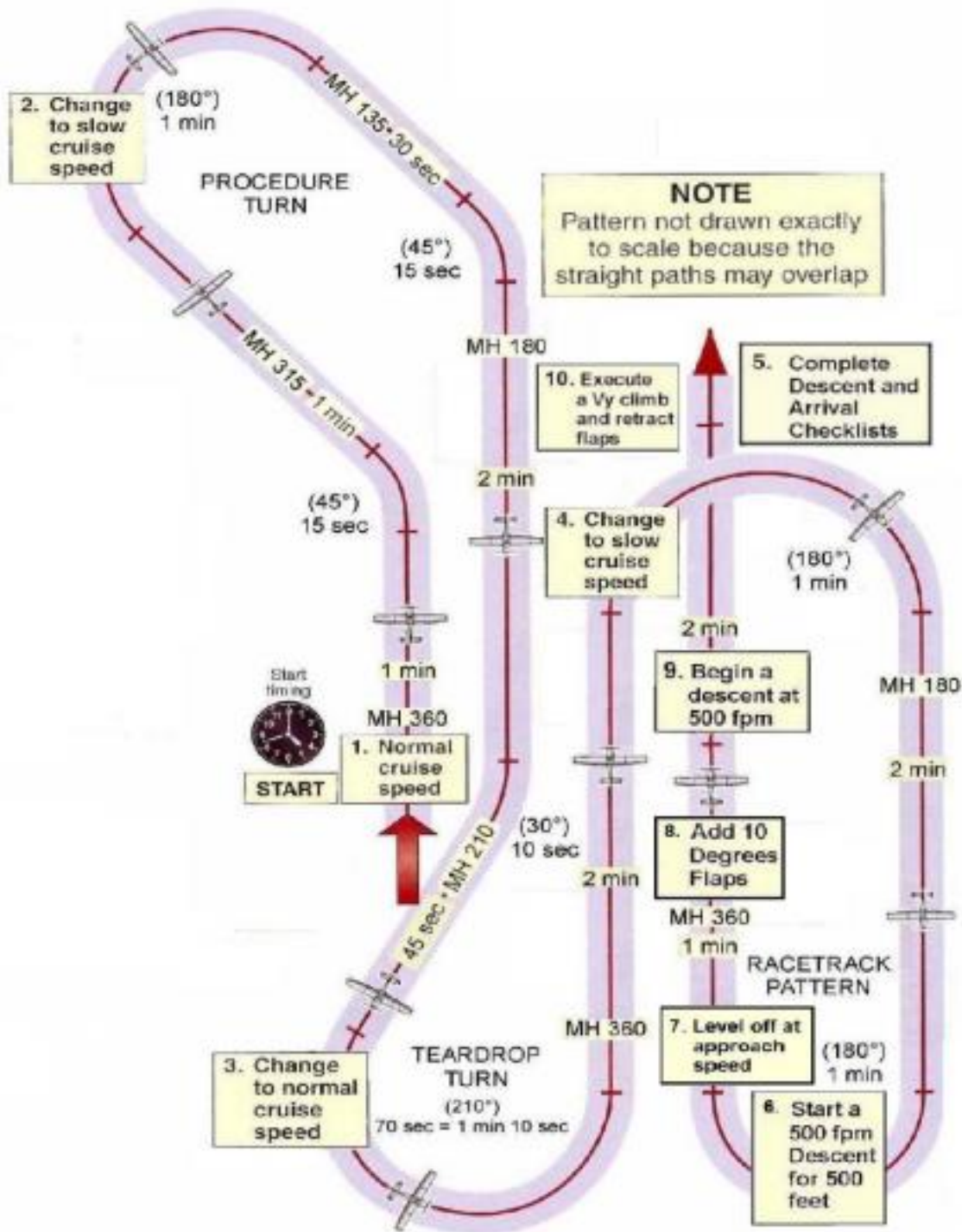
Pattern D



Lesson Plans

Instrument Rating Course

Pattern F



Lesson Plans

Instrument Rating Course

Vertical S-1

